LeWM checkpoint reproduction

Results: checking the foundation before building on it

We evaluated the released LeWorldModel (LeWM) checkpoints to check whether they provide a competent starting point for our planning and safety research. Weights stayed frozen, and control used the original Cross-Entropy Method (CEM) planner. There was no retraining, Anchored Planning or safety adapter.

What we tested. Each start and goal came from the same recorded trajectory, with the goal **25 environment actions ahead**. The controller had **50 actions** to reach it. We tested 50 cases per environment for seeds 42, 43 and 44: 600 main episodes. Seed 42 was the primary paper comparison; the other seeds were robustness repeats using the same checkpoint.

Environment	Paper	Primary (42)	Repeats (43 / 44)
TwoRoom	87%	90% (45/50)	74% / 90%
Reacher	86%	64% (32/50)	68% / 86%
PushT	96%	96% (48/50)	88% / 88%
OGBench Cube	74%	72% (36/50)	84% / 64%

CEM used 300 candidates, 30 elites and a 25-action plan before replanning. The main settings used 10 CEM iterations for TwoRoom, Reacher and Cube, and 30 for PushT. A separate, predeclared 30-iteration check reused the seed-42 cases for the other three tasks: TwoRoom 84%, Reacher 78% and Cube 68%. These 150 extra episodes brought the total to **750 evaluations**.

What we learned. Before running, we allowed a primary result up to 10 percentage points below the paper as a practical competence check. **TwoRoom, PushT and Cube met this tolerance; Reacher did not.** The extra-iteration result does not replace Reacher's failed main comparison. Across all three main seeds, success was 84.7%, 72.7%, 90.7% and 73.3%, respectively. Variation across cases remains substantial; this is not proof of equivalence to the paper or a replication across training seeds.

Checks and limitations. All 15 evaluation panels passed the evidence audit, covering 37,500 controller actions. Exact cases, outcomes, configuration, model queries and uncertainty intervals are saved, with verified backups in our private Hugging Face dataset. Runtime and paper/code differences are documented, including rendering, goal indexing and checkpoint history length. Engineering used another 710 actions. Ten raw trace rows from an early logging failure were lost; their action count is preserved and the limitation is recorded.

The result gives us a tested **nearby-goal baseline for three environments**. Reacher's shortfall remains unresolved. Safety, longer-horizon control and any benefit from our adaptations still require separate experiments.

Completed 4 November 2026. Frozen checkpoints, native CEM, fixed evaluation panels.